

Big Data - Assessment 1

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Confirmation from Power BI

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Analysis from the chart

Confirmation from Power BI

Conclusion for analysis

ASIN & SKU

Ship city & Date

Promotion ID & Date

Category & Amount

Status & Courier status & Fulfilment

Status & Quantity

ASIN and Style

Fulfilment & Ship service level

Implement Classification to the dataset

View the raw result on GitHub

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Recall (TP Rate)

Precision

F-Measure

Case analysis

Evaluate negative effect

Nature of out case - Selling clothes on Amazon

Analyze acceptance of negative effects relying on the nature

Conclusion

About the Project and Dataset Preparation

This project is about: Data Analysis and Classification for E-Commerce Sales Dataset

Dataset provides detailed insights into Amazon sales data, including SKU Code, Design Number, Stock, Category, Size and Color, to help optimize product profitability

1. <https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data>

Workflow of this project

This project is processed basically following the sequenced operations

1. Choose the target dataset from Kaggle

<https://www.kaggle.com/>

2. Original dataset need to be modified to be suitable for later data mining

modifications include:

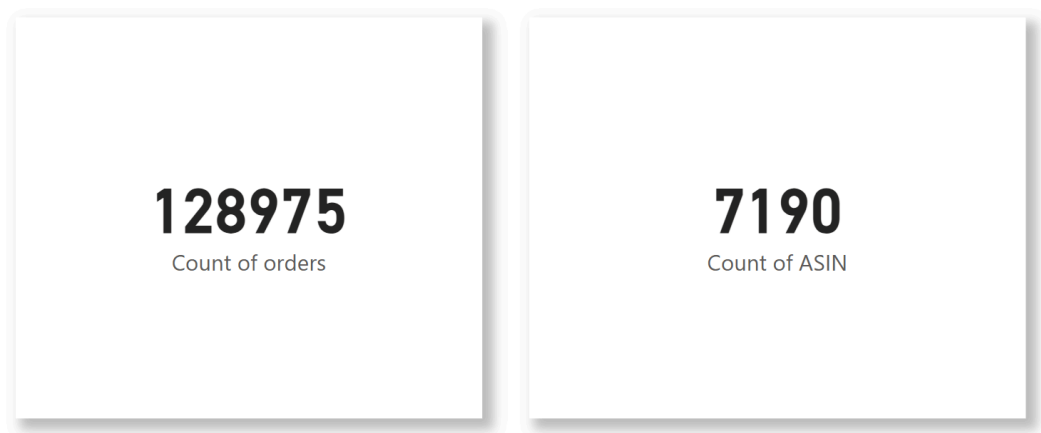
- a. Get rid of missing values
- b. Remove outliers
- c. Remove insignificant attributes

3. Use Weka for data mining and find correlations of dataset
 - a. Visualize the dataset inside scatter plot
4. If necessary, use Power BI to prove and reevaluate the hypothesis from Weka
5. Utilize four classifiers and get training result of dataset

Description of selected dataset

1. Feature of Amazon orders dataset

- Using Power BI and fetch 128975 records with 7190 distinct ASINs from the dataset



2. Purpose and outcome analyzing the dataset

- This dataset include details of orders on Amazon basically from India. Analyzing this dataset, we can estimate **possible correlations among these attributes**
- → Sort out is there certain dates for people living in certain regions when they get crazy desire to purchase on Amazon (like Western countries During Black Friday)
- Therefore we could help Amazon sellers find our **potential strategies** with evidence to enhance profitability from their business

Reduction as well as preparation of attributes

1. Originally present attributes

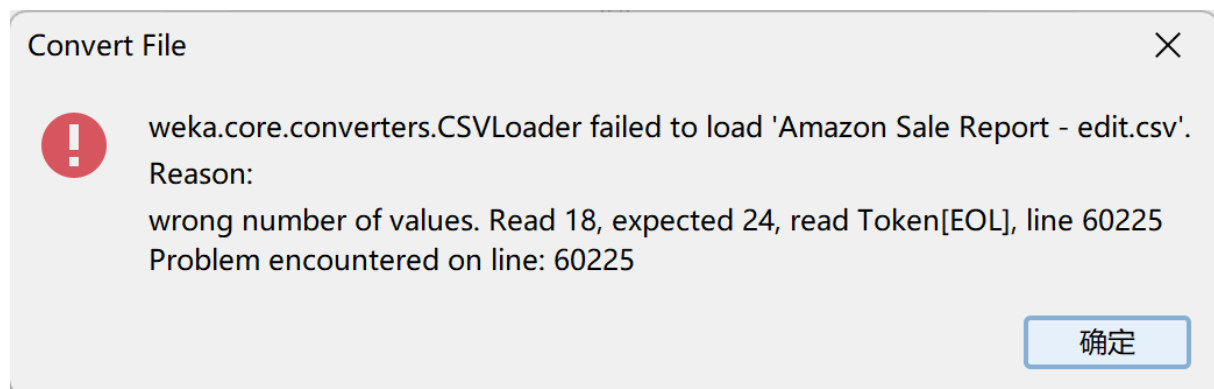
- Their are 24 types of attributes in the original data set including {index, Order ID, Date, Status, Fulfilment, Sales Channel, ship-service-level, Style, SKU, Category, Size, ASIN, Courier Status, Qty, currency, Amount, ship-city, ship-state, ship-postal-code, ship-country, promotion-ids, B2B, fulfilled-by, Unnamed: 22}
- We don't need all of those attributes since some of them are less representative indicate certain traits of Sales analysis, therefore reduction to these attributes is necessary

2. Reduced attributes with descriptions for types and business aims

- **Category** : Type of product. (String)
 - →
- **Size** : Size of the product. (String)
- **Date** : Date of the sale. (Date)
- **Status** : Status of the sale. (String)
- **Fulfilment** : Method of fulfilment. (String)
- **Style** : Style of the product. (String)
- **SKU** : Stock Keeping Unit. (String)
- **ASIN** : Amazon Standard Identification Number. (String)
- **Courier Status**: Status of the courier. (String)
- **Qty** : Quantity of the product. (Integer)
- **Amount** : Amount of the sale. (Float)
- **B2B** : Business to business sale. (Boolean)
- **Currency** : The currency used for the sale. (String)

Convert CSV file into ARFF file

An error occurs that seems because of invalid syntax `'`



Therefore, for each string containing syntax `'`, I added `"` on its both sides

Near Laurel's international school,

>>>>>>

"Near Laurel's international school"

Then, the CSV successfully converted into ARFF and be loaded onto Weka

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation
Relation: Amazon Sale Report - edit
Instances: 128975
Attributes: 24
Sum of weights: 128975

Attributes
All None Invert Pattern

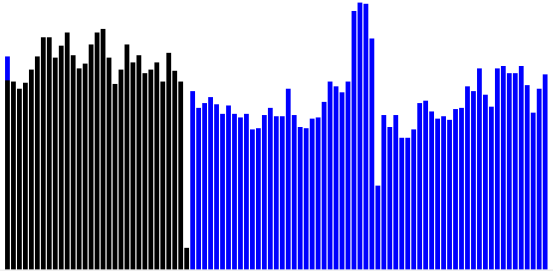
No.	Name
1	☐ index
2	☐ Order ID
3	☒ Date
4	☐ Status
5	☐ Fulfilment
6	☐ Sales Channel
7	☐ ship-service-level
8	☐ Style
9	☐ SKU
10	☐ Category
11	☐ Size
12	☐ ASIN
13	☐ Courier Status
14	☐ Qty
15	☐ currency
16	☐ Amount
17	☐ ship-city
18	☐ ship-state
19	☐ ship-country

Remove

Selected attribute
Name: Date
Missing: 0 (0%)
Distinct: 91
Type: Nominal
Unique: 0 (0%)

No.	Label	Count	Weight
1	04-30-22	1669	1669
2	04-29-22	1466	1466
3	04-28-22	1416	1416
4	04-27-22	1459	1459
5	04-26-22	1563	1563
6	04-25-22	1665	1665
7	04-24-22	1810	1810
8	04-23-22	1816	1816
9	04-22-22	1657	1657
10	04-21-22	1746	1746

Class: Unnamed: 22 (Nom) Visualize All



Status: OK Log x 0

Remove missing values and outliers

Relation: Amazon Sale Report - edit													16: Amount	17: ship-city	18: ship-state	19: ship-postal-code	20:
8: Style	9: SKU	10: Category	11: Size	12: ASIN	13: Courier Status	14: Qty	15: currency										
Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Numeric	Nominal										
JNE38...	JNE38...	Western Dr...	XXL	B09TH5...		0.0	INR	734.29	VIJAYAWA...	ANDHRA P...	520012.0	IN					
J0381	J0381...	Set	3XL	B09M6...		0.0	INR	1023.21	VIJAYAWA...	ANDHRA P...	520012.0	IN					
JNE37...	JNE37...	Western Dr...	3XL	B09SDY...		0.0	INR	689.52	VIJAYAWA...	ANDHRA P...	520012.0	IN					
PJNE3...	PJNE3...	kurta	6XL	B09LD4...	Shipped	1.0	INR	486.0	BENGALURU	KARNATAKA	560036.0	IN					
PJNE2...	PJNE2...	kurta	6XL	B09LD4...	Shipped	1.0	INR	431.0	BENGALURU	KARNATAKA	560036.0	IN					
JNE33...	JNE33...	kurta	L	B081X4...	Shipped	1.0	INR	471.0	MHOW	MADHYA P...	453441.0	IN					
SET227	SET22...	Set	L	B08P1D...	Shipped	1.0	INR	571.0	Allahabad	UTTAR PRA...	211016.0	IN					
SET331	SET33...	Set	XXL	B09NQ...	Shipped	1.0	INR	597.0	ERNAKUL...	KERALA	682020.0	IN					
JNE32...	JNE32...	kurta	M	B07R5X...	Unshipped	1.0	INR	348.0	HYDERABAD	TELANGANA	500015.0	IN					
JNE36...	JNE36...	kurta	M	B091Q8...	Shipped	1.0	INR	318.0	HOSKOTE	KARNATAKA	562114.0	IN					
JNE36...	JNE36...	Top	L	B0943J...		0.0			HOSKOTE	KARNATAKA	562114.0	IN					
PJNE3...	PJNE3...	kurta	4XL	B09M7...	Shipped	1.0	INR	692.0	BAGALKOT	KARNATAKA	587101.0	IN					
JNE37...	JNE37...	kurta	3XL	B09K3Z...	Shipped	1.0	INR	362.0	HYDERABAD	TELANGANA	500016.0	IN					
JNE37...	JNE37...	kurta	3XL	B09K3T...	Shipped	1.0	INR	345.0	HYDERABAD	TELANGANA	500016.0	IN					
JNE38...	JNE38...	Western Dr...	XXL	B09RK6...	Shipped	1.0	INR	677.0	RAJPUR S...	WEST BENG...	700148.0	IN					
J0135	J0135...	Set	XL	B08QGJ...	Shipped	1.0	INR	591.0	JAMMIKU...	TELANGANA	505122.0	IN					
SET293	SET29...	Set	3XL	B09K3C...	Shipped	1.0	INR	736.0	MUMBAI	MAHARASH...	400018.0	IN					
SET377	SET37...	Set	XXL	B09TZT...	Shipped	1.0	INR	1238.0	MATHURA	UTTAR PRA...	281003.0	IN					
JNE37...	JNE37...	kurta	M	B09K3X...	Shipped	1.0	INR	499.0	BENGALURU	KARNATAKA	560068.0	IN					
J0094	J0094...	kurta	M	B089G1...	Shipped	1.0	INR	568.0	BENGALURU	KARNATAKA	560068.0	IN					
J0012	J0012...	Set	XL	B0894...	Shipped	1.0	INR	1140.0	THANE	MAHARASH...	400612.0	IN					
SET331	SET33...	Set	M	B09NQ...	Unshipped	1.0	INR	597.0	THANE	MAHARASH...	400608.0	IN					
J0119	J0119...	Top	M	B08RY...	Shipped	1.0	INR	0.0	Bangalore	KARNATAKA	560016.0	IN					

- Blank value on currency and amount do not make sense on real case of sales on Amazon, therefore, they are to be removed from the dataset

Filter

Choose RemoveWithValues -S 0.0 -C 15 -L 1 -M

- Index and Order ID are unique or auto-increasing, it's unnecessary to include these attributes since they are not considered as features related to the case itself, furthermore it could be overfitting and does not contribute to data prediction

No. Name

1 ☒ index

2 ☒ Order ID

3 ☐ Date

4 ☐ Status

5 ☐ Fulfilment

6 ☐ Sales Channel

7 ☐ ship-service-level

8 ☐ Style

9 ☐ SKU

10 ☐ Category

11 ☐ Size

12 ☐ ASIN

13 ☐ Courier Status

14 ☐ Qty

15 ☐ currency

16 ☐ Amount

17 ☐ ship-city

18 ☐ ship-state

19 ☐ ship-postal-code

20 ☐ ship-country

21 ☐ promotion-ids

22 ☐ B2B

23 ☐ fulfilled-by

24 ☐ Unnamed: 22

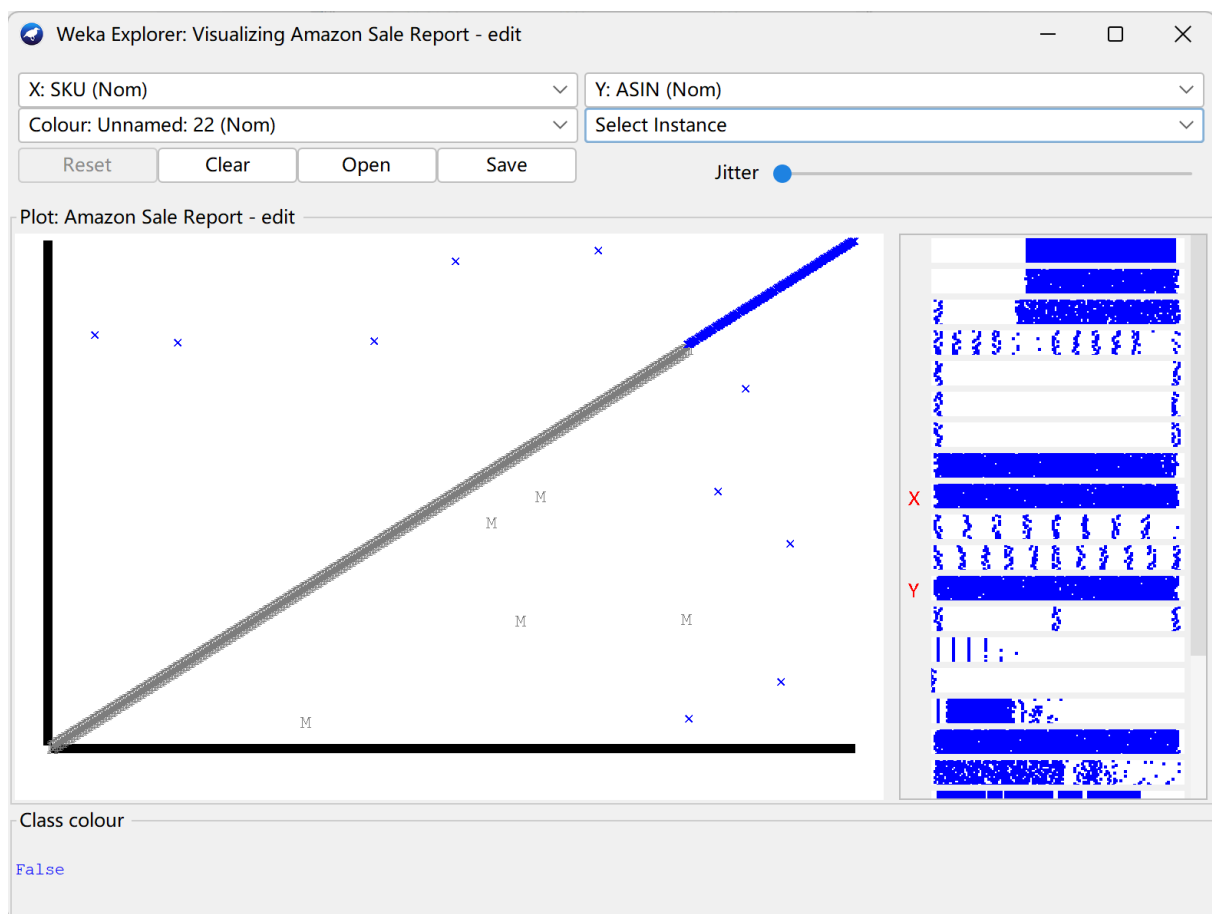
Remove

Data visualization and analysis

- ASIN & SKU - One ASIN related to one SKU?

- Ship city & Date - Gap in Orders frequency between different cities
- Promotion ID & Date
- Category & Amount
- Status & Courier status
- Status & Quantity
- Status & Fulfilment
- Fulfilment & Ship service level

1. Correlation between ASIN and SKU



One-to-one direct correspondence

- This chart illustrates values in plot forms a straight diagonal line, which indicate that at most every SKU has its own ASIN respectively
- Their Correlation as a potential **one-to-one direct** correspondence.

Outlier to one-to-one direct correspondence

- However there are also points scatter around this line, which means there are couple set of SKU and ASIN that do not strictly follow one-to-one correspondence.
- One SKU may map to **Multiple** ASINs or vice versa, while this situation is **minority**.

Business implication

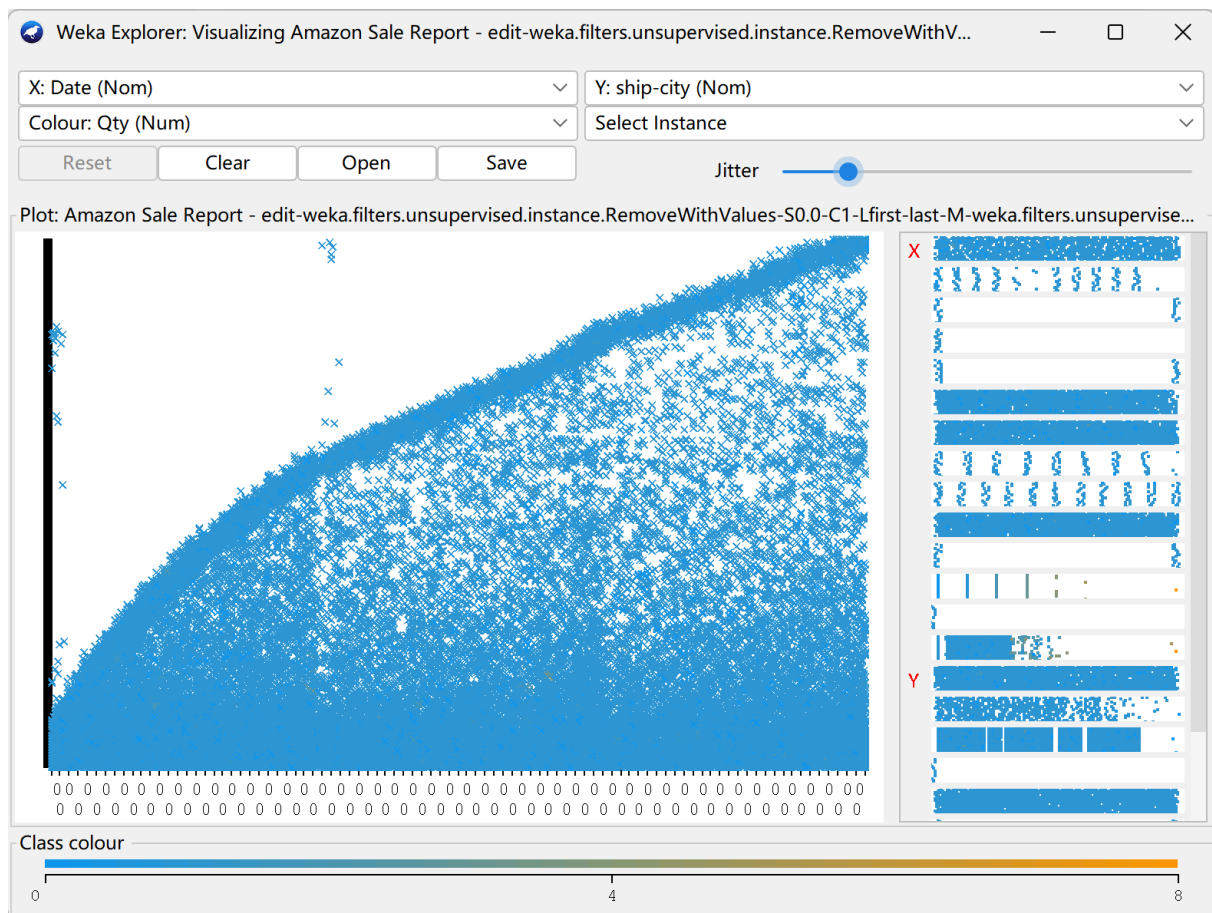
- Existence of outliers may because Amazon distribute different ASINs for a single SKU in different region (assumption).

Confirmation from Power BI

- One-to-many correlation existed as minority as it assumed by the scatter plot where SKU and ASIN are mostly following one-to-one direct Correlation with a few exceptions

SKU	count of ASIN	ASINs
J0027-SET-XS	2	B08B3YZN3J, B08B4T6KJK
J0091-TP-XL	2	B092D2LWSQ, B092ZXCCNJ
J0157-DR-S	2	B0982ZDZ8K, B098P3LXT1
J0212-DR-L	2	B0982Z8HPJ, B098P4Z9WZ
JNE3713-TP-N-XXL	2	B091GV1PQR, B08ZHGDW8
AN201-RED-M	1	B091GP22TP
AN201-RED-XL	1	B091GLZF98
AN201-RED-XXL	1	B091GP61BH
AN202-ORANGE-M	1	B091GMKCQP
AN202-ORANGE-S	1	B091GP4R8Y
AN202-ORANGE-XXL	1	B091GK5Y64
AN203-MAROON-XXL	1	B091GPQB62

2. Correlation between Ship city and Date



Rough one-to-one direct correspondence

- This chart illustrates a rough diagonal line when upscaling jitter
- **One-to-one correlation** between city and a **period of dates**
- → There are periods for cities with frequent sale activities but not consistently keep the frequency over all.

For instance, customers living in A make unusually frequent orders during March and others living in B do so during April. The regional pattern may be led by **regional promotion**.

Gap in density

- The density for ship-city over time is not flat - some area is more dense than others.
- There are two features about density
 - Extremely low density at top-left area
 - high density at bottom-right
- → Indicate Huge regional gap in sale frequency - some city only have few purchases while some have high loyal customers.

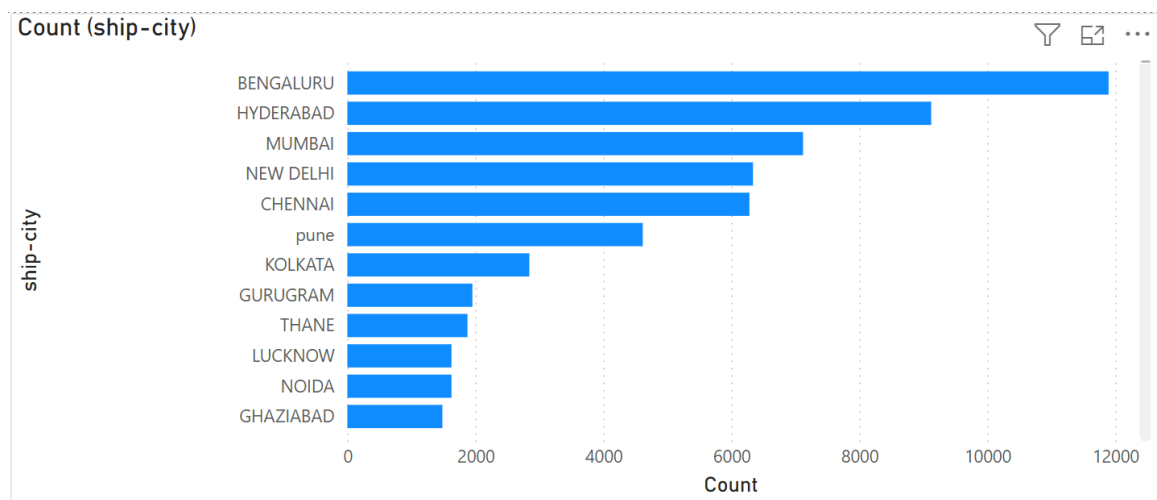
Several potential reasons

- The high **regional gap** in economy and population in India (most data taken from India)
- Regional gap in acceptance to Amazon (residents in different cities tends to different shopping platform)

Commonly, former explanation is more realistic.

Confirmation from Power BI

- **Regional Gap**



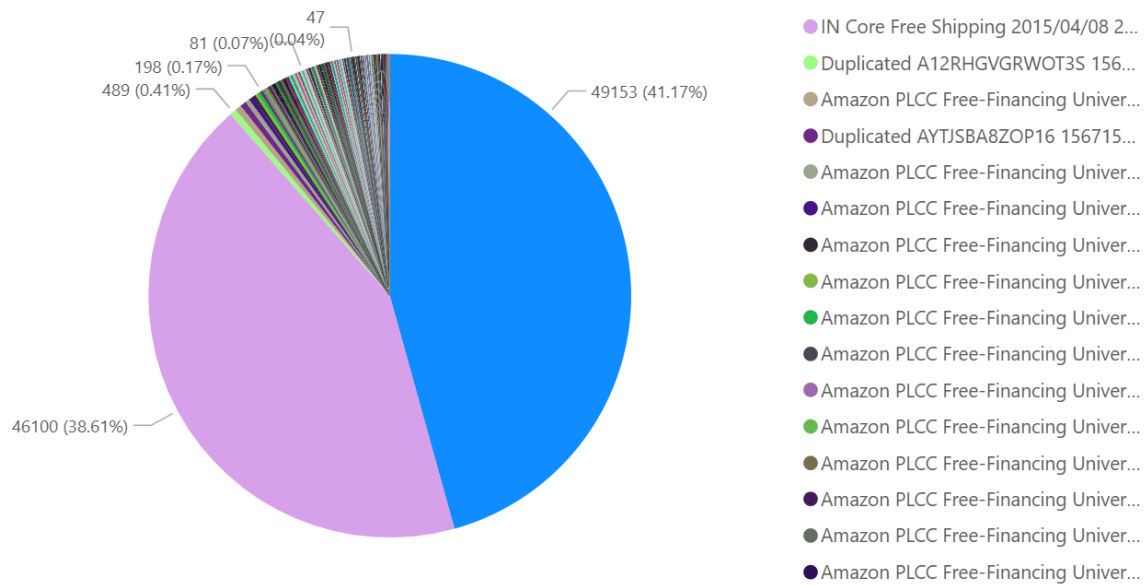
- This chart shows the regional gap in orders count that Bengaluru get 11898 orders while the city with the no.12 orders - Ghaziabad get 1485.



According to Wikipedia, Bengaluru and Ghaziabad have respectively 8448K and 1729K (data from 2011), that means each resident in Bengaluru make 60% more Amazon purchase than one in Ghaziabad.

3. Correlation between Promotion ID and Date

Count (promotion-ids)



Additionally, with the pie chart created by Power BI which shows the empty value share the biggest part with 41% and subsequently followed by **IN Core Free Shipping 2015/04/08 23-48-5-108** with 39%, which must be the main part of the lying parallax line of values.

4. Correlation between Category and amount over period

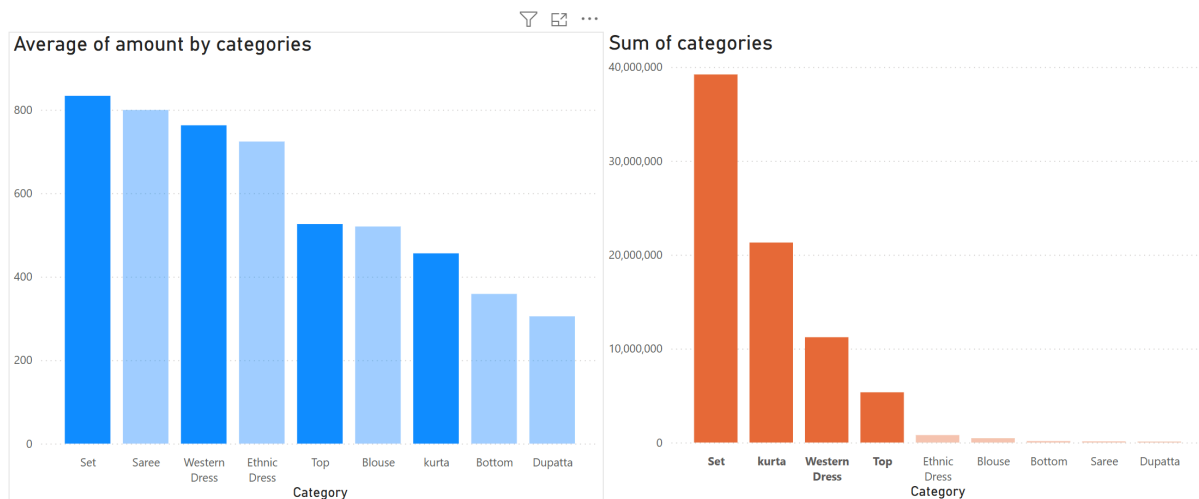


Analysis from the chart

- It is evident that the diagram is divided into parts of area where there is majority of certain colour
- Generally, blue colour occupied the upper area which followed by green, light blue and red from top to bottom
- Goods with category of **Set** take place most often and with relatively high prices, which followed by the green one - **Western Dress**, then the light blue one - **Top**, as well as the red one - **Kurta**, following from costly to cheap.
- It is also noticeable that the rank is general but not precise, for instance the price of the green one - **Western Dress** could be more expensive than **Set**.

Furthermore, the proportions for general amount of category do not seem to alter much over period, which indicate the customers in the survey have quite stable need for each categories of good over time.

Confirmation from Power BI

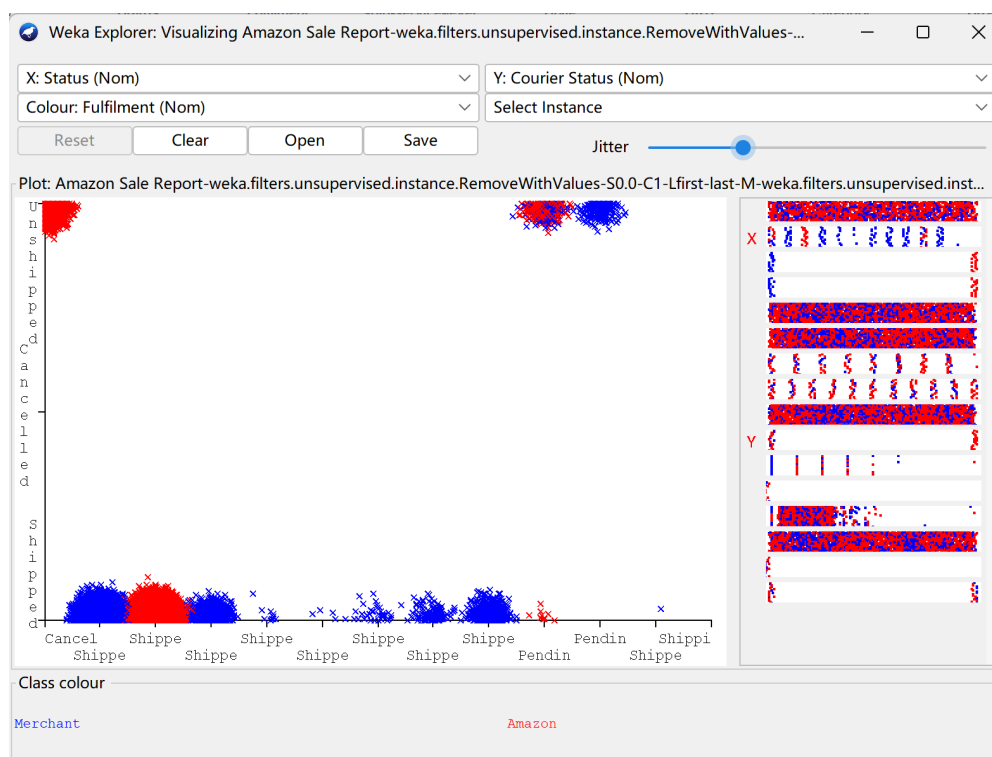


These two charts illustrate count and average of amount for every categories. By selecting top 4 often categories at the right chart which is **Set** , **Western Dress** , **Top** and lastly **Kurta** .

It is evident as the highlighted categories at the left chart that the average amount for these categories are ranked as mostly the same as they were analyzed from Weka.

By the way, it is interesting to find that both the average amount and count for good of **Set** are highest. That is why it is obviously illustrated with the largest area of 'blue' as it represents that goods of **Set** are the best-selling category over period.

5. Correlation between Status and Courier Status



Analysis from the chart

The chart illustrate correlation between Status and Courier status, which respectively related to annotation to status of orders and status of fulfilment.

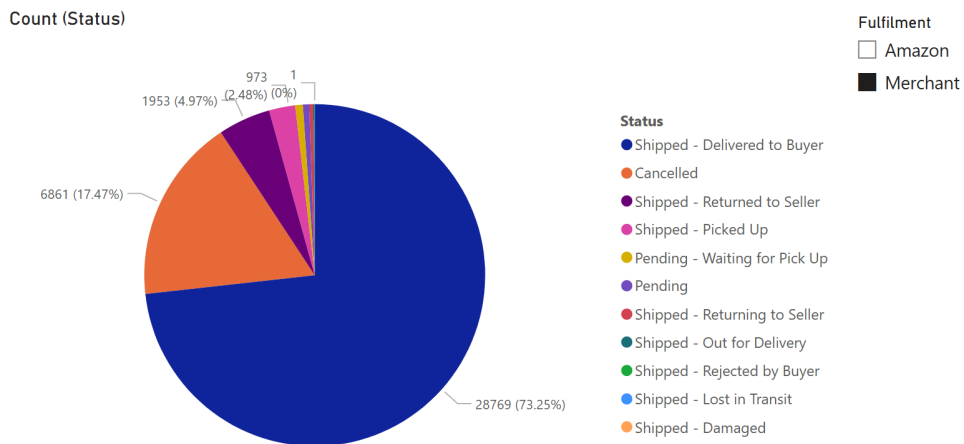
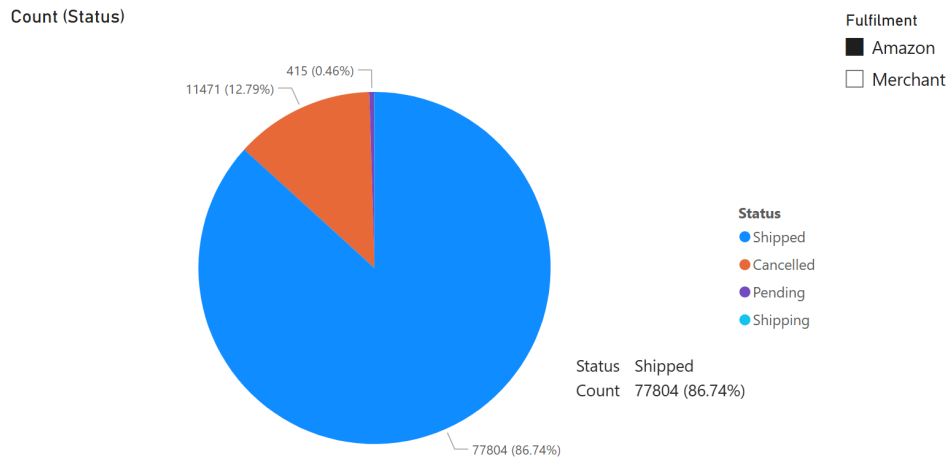
They are both nominal with a few elements so I upscale jitter to get better vision on scatter plot, where they follow a clear relations:

- If an order has courier status as **Cancelled** , the status is to be **Unshipped** and it is to be fulfilled by Amazon
- Those with Amazon fulfilment only have status of **Shipped** , **Cancelled** and **Pending** , while those with Merchant fulfilment have more detailed status shipped status instead of **Shipped** and other to **Pending** , there is **Pending - Waiting for Pick Up** in status.
- There are no **Cancelled** in Courier status, this is because those with it get blank value in currency and amount which have been removed during data preparation stage.
- If an order has **Pending** as its status, it often means **Unshipped** as its courier status. However, this is not an exact correlation - sometime **Pending** status together with **Shipped** as courier status.

Confirmation from Excel

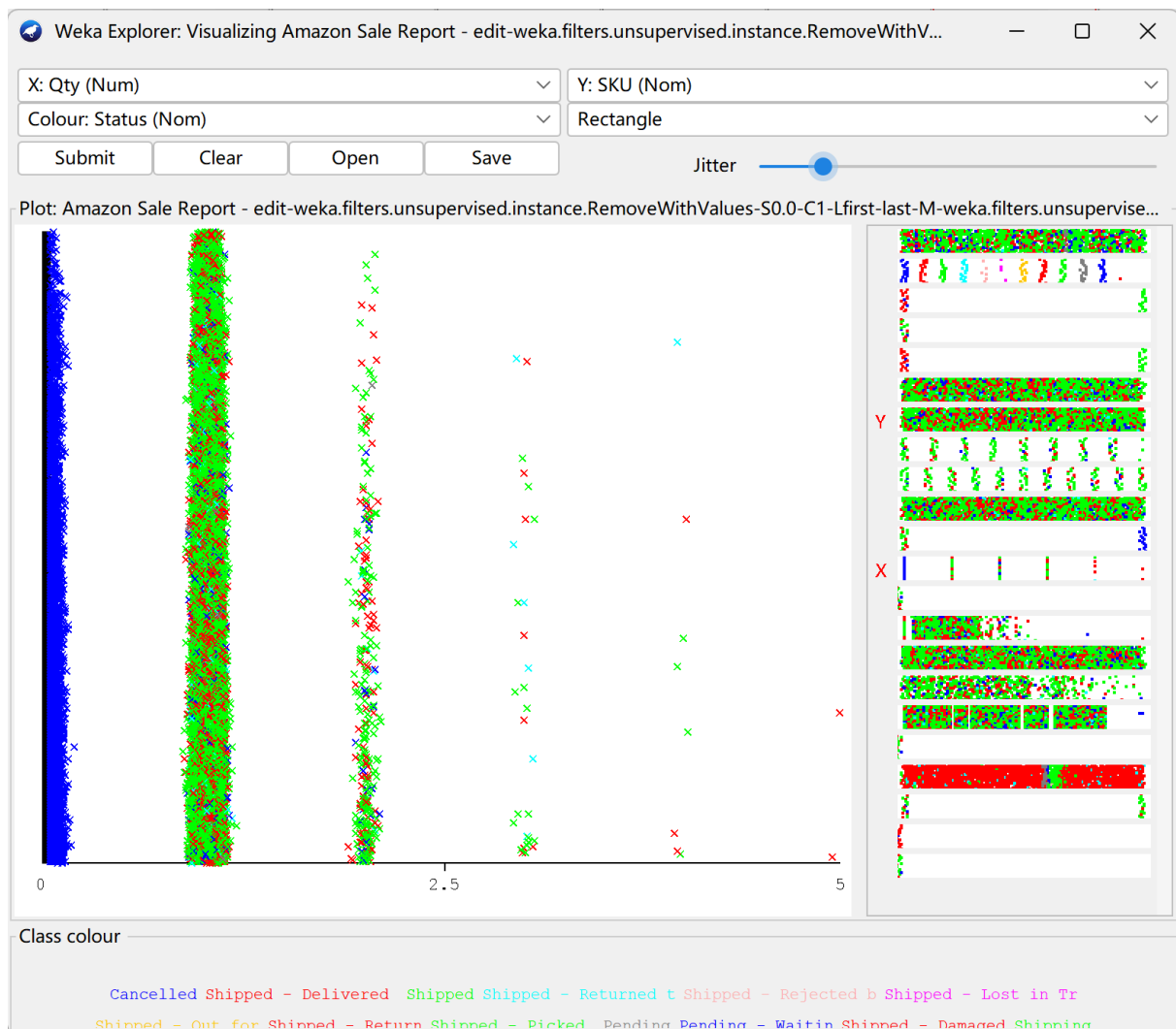
B	K	L	M	N
Status	Courier Status	Qty	currency	Amount
Cancelled		0	INR	647.62
Cancelled		0	INR	753.33
Cancelled	Cancelled	0		
Cancelled		0	INR	570.48
Cancelled		0		
Cancelled	Cancelled	0		
Cancelled		0	INR	1105.36
Cancelled	Cancelled	0		
Cancelled	Cancelled	0		
Cancelled	Cancelled	0		
Cancelled	Unshipped	1	INR	791
Cancelled	Unshipped	1	INR	807
Cancelled	Unshipped	1	INR	807
Cancelled	Cancelled	0		
Cancelled	Cancelled	0		
Cancelled	Cancelled	0		
Cancelled	Cancelled	0		

Records with CANCELLED courier status confirmed to get empty **currency** and **amount**



These two pie charts prove Amazon and Merchant fulfilments get exclusive approaches to mark status - Amazon get simpler divisions of status while Merchant ones go with more detailed categories.

6. Correlation between Status and Quantity

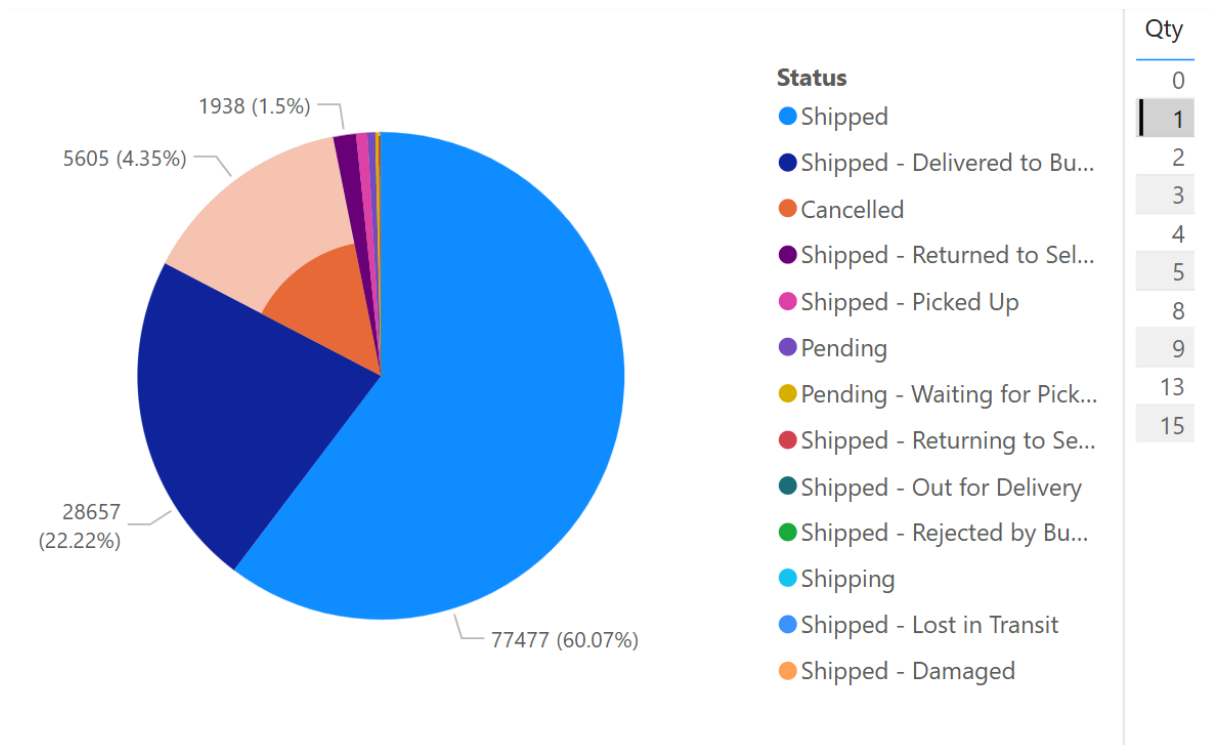


Analysis from the chart

It is evidence that the variety of SKU shrinks dramatically as the quantity grows - most purchase involves single product, while there a considerable number of cancelled records all of which get 0 in quantity.

It is clear that orders with **Shipped** and **Shipped - Delivered** get the largest share on non-cancelled orders due to the dominating areas of green and red, which means most orders was successfully delivered with several goods returned. Notably, orders with one Qty share the most proportion among all and cover almost every SKU.

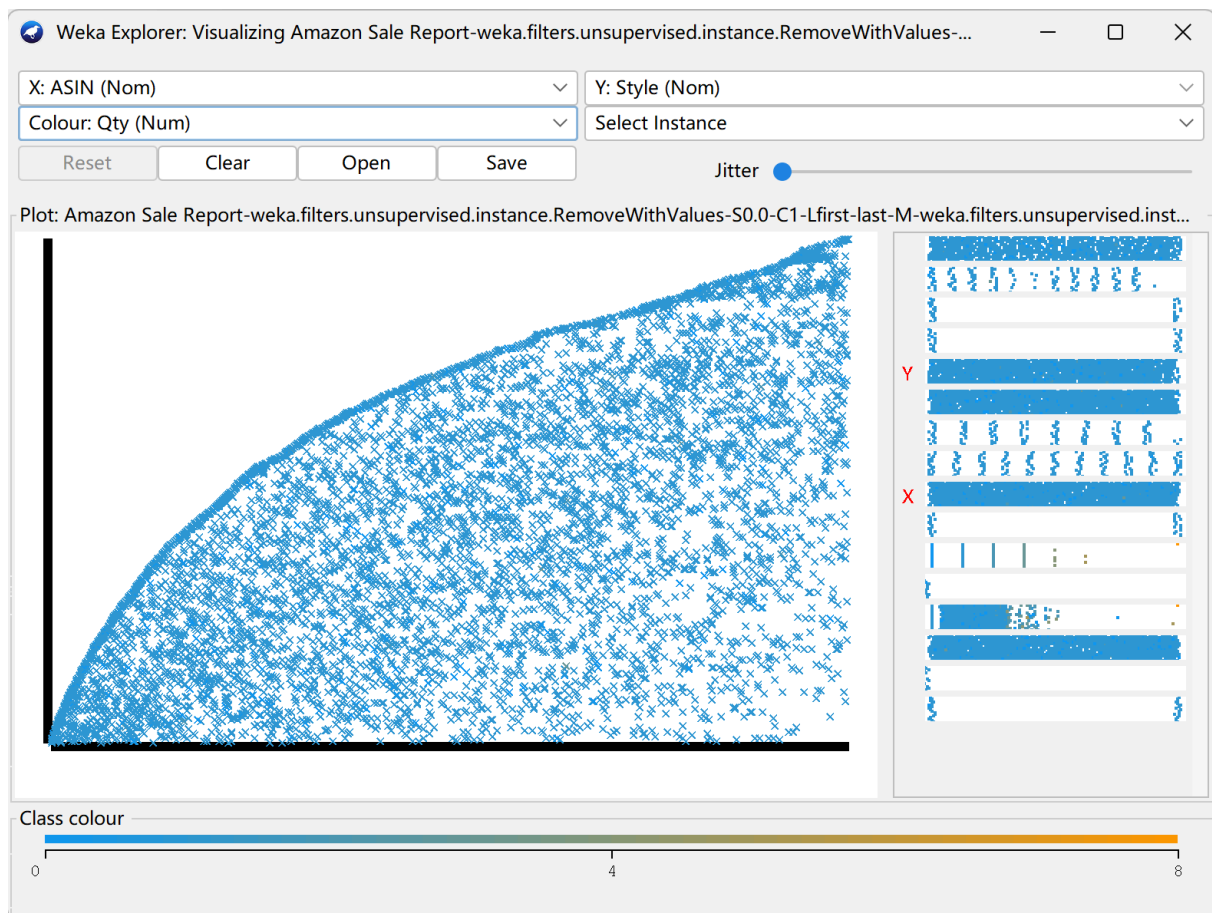
Confirmation from Power BI



The pie chart shows the share of all statuses filtered by condition with one Qty. Single Qty orders are the majority for every sort of status, except from **Cancelled** which mainly occur in those with 0 Qty.

Shipped and **Shipped - Delivered to Buyer** hold the largest share at respectively 60.07% and 22.22%, which justifying our assumption from Weka

7. Correlation between ASIN and Style



This chart illustrate two features - high density at bottom-right and a rough diagonal dense line

Dense at bottom-right

- Points mostly scattered at bottom-right means ASIN is more various than Style, which mean one Style could be mapped to several ASINs

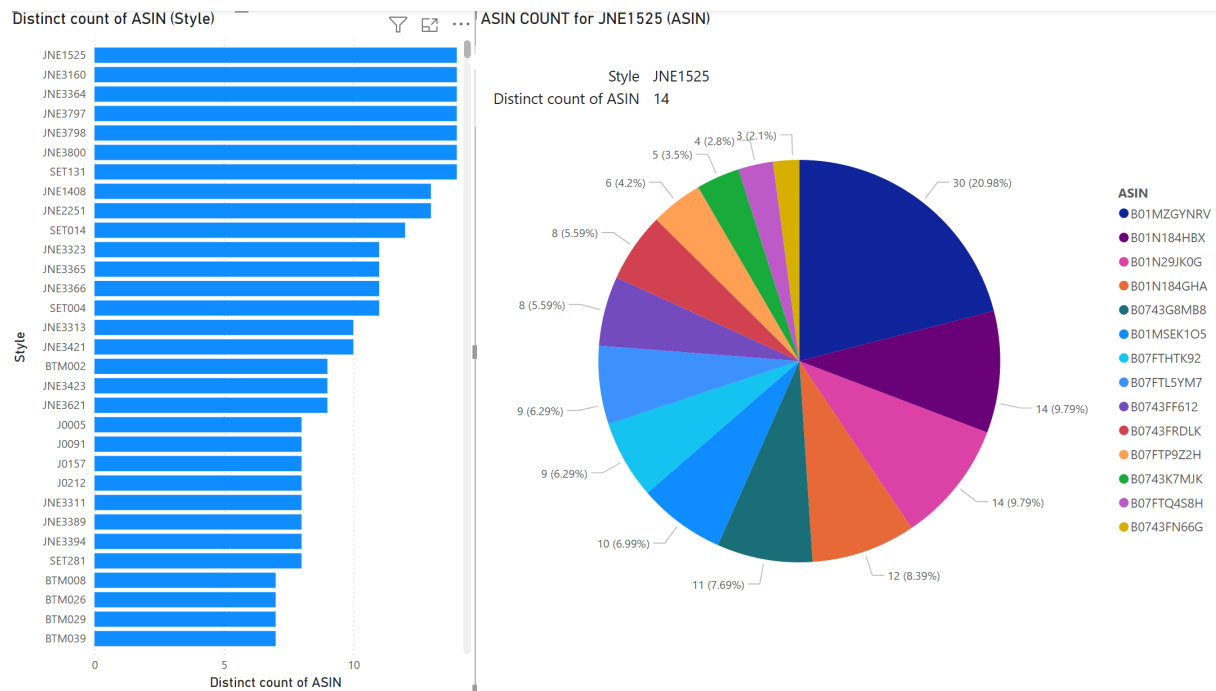
Dense diagonal line

- There are many combinations of ASIN and Style with positive relation, which means there are always ASIN more popular than others that is sharing the same style.

For instance, style A get ASINs of 'a', 'b', 'c' to combine with, and 'a' is much popular than 'b' and 'c'.

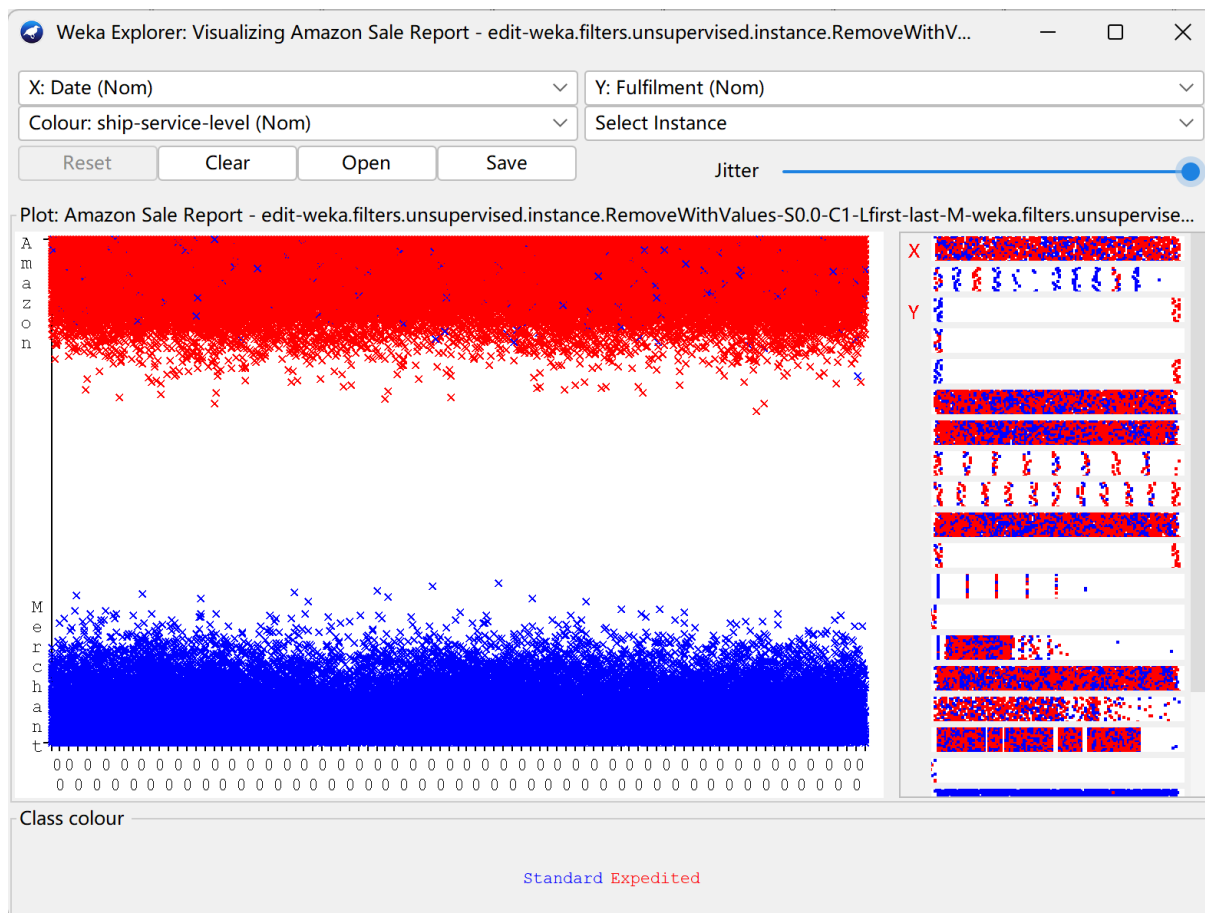
This reveal a business phenomenon that customers tend to have similar preference. Shopkeeper should pay attention to import more models of more popular ones to reduce stock redundancy.

Confirmation from Power BI



It is evident shows on left chart that there are styles with many possible ASINs (for instance JNE1525 has 14 kinds of ASINs). While the pie chart on the right shows proportion of count of ASINs of JNE1525, where ASIN of **B01MZGYNRV** is clearly the most popular ASIN (30 purchases in total 143)

8. Correlation between Fulfilment and Ship-Service-Level



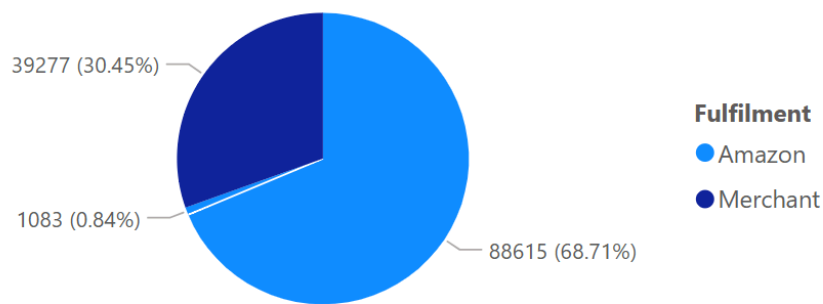
Analysis from the chart

This chart shows two lines with red and blue for each line. For prominent graph, Jitter is adjusted at highest. Then it is clear that fulfilment almost dictate the ship service level, where Amazon fulfilment provide Expedited service which is used in most Amazon orders with a few exception. In contract, order with Merchant fulfilment exclusively used standard ship level with no exception according to chart where no red point seen among the blue area.

Confirmation from Power BI

This pie chart proves our research at Weka - Amazon monopolize the **Expedited** Fulfilment, while some Amazon order still use **Standard**

Count (Fulfilment and ship-service-level)



Fulfilment	Count
☐ Amazon	
Expedited	88615
Standard	1083
☐ Merchant	
Standard	39277

Conclusion for analysis

ASIN & SKU

- Mostly one-to-one direct correlation
- One SKU may mapped to several ASIN

Ship city & Date

- Huge distinction of number of orders among cities
- Number of orders Increase in certain cities during certain period → Regional promotion expected

Promotion ID & Date

- Most Promotion ID applied during a period and expired after then
- Some Promotion ID could be applied almost everyday

Category & Amount

- There are ranges of price related to category of good

Status & Courier status & Fulfilment

- Elements of status depends on the fulfilment of order
- One courier status relates to several potential statuses
- Most orders were shipped

Status & Quantity

- Most orders go with one quantity of good

- All orders with zero quantity is cancelled
- Most orders - no matter its quantity - were shipped

ASIN and Style

- One style corresponds to multiple ASINs
- For ASINs with the same style, there are ASINs that are obviously more popular than others

Fulfilment & Ship service level

- Expedited ship is exclusively available in Amazon fulfilment
- Those with Merchant fulfilment are utilized Standard ship
- Most orders with Amazon fulfilment utilized Expedited ship

Implement Classification to the dataset

Setting the predicting attribute to **Status** as its positive value as **Cancelled**

It was applied **Four kinds of classifications** each of which being diverted to **Two size for training set** - J48, simple cart, native bayes, KNN which varied by setting different training set size (60% and 80%)

View the raw result on GitHub

- <https://github.com/licwan62/DataAnalysis-AmazonSales/tree/main>

Output of all classifications

1. Result by Detailed Accuracy and Confusion Matrix

=== Detailed Accuracy By Class ===									
	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.994	0.001	0.994	0.994	0.994	0.993	1.000	0.998	Cancelled
	0.996	0.024	0.928	0.996	0.961	0.949	0.989	0.936	Shipped - Delivered to Buyer
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Shipped
	0.000	0.000	?	0.000	?	?	0.890	0.069	Shipped - Returned to Seller
	0.000	0.000	?	0.000	?	?	0.491	0.000	Shipped - Rejected by Buyer
	0.000	0.000	?	0.000	?	?	0.994	0.004	Shipped - Lost in Transit
	0.000	0.000	?	0.000	?	?	0.925	0.018	Shipped - Out for Delivery
	0.000	0.000	?	0.000	?	?	0.953	0.052	Shipped - Returning to Seller
	0.845	0.001	0.834	0.845	0.839	0.838	0.985	0.896	Shipped - Picked Up
	0.820	0.001	0.882	0.820	0.850	0.850	0.973	0.894	Pending
	1.000	0.000	0.865	1.000	0.928	0.930	1.000	0.886	Pending - Waiting for Pick Up
	0.000	0.000	?	0.000	?	?	0.500	0.000	Shipped - Damaged
	?	0.000	?	?	?	?	?	?	Shipping
Weighted Avg.	0.979	0.006	?	0.979	?	?	0.995	0.967	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	h	i	j	k	l	m	
a	4228	0	0	0	0	0	0	0	0	27	0	0	0	<-- classified as
b	0	11370	0	0	0	0	0	0	49	0	0	0	0	a = Cancelled
c	0	0	31234	0	0	0	0	0	0	0	0	0	0	b = Shipped - Delivered to Buyer
d	0	754	0	0	0	0	0	0	0	0	0	0	0	c = Shipped
e	0	4	0	0	0	0	0	0	0	0	0	0	0	d = Shipped - Returned to Seller
f	0	2	0	0	0	0	0	0	0	0	0	0	0	e = Shipped - Rejected by Buyer
g	0	9	0	0	0	0	0	0	6	0	0	0	0	f = Shipped - Lost in Transit
h	0	52	0	0	0	0	0	0	8	0	0	0	0	g = Shipped - Out for Delivery
i	0	58	0	0	0	0	0	0	316	0	0	0	0	h = Shipped - Returning to Seller
j	25	0	2	0	0	0	0	0	0	201	17	0	0	i = Shipped - Picked Up
k	0	0	0	0	0	0	0	0	0	0	109	0	0	j = Pending
l	0	1	0	0	0	0	0	0	0	0	0	0	0	k = Pending - Waiting for Pick Up
m	0	0	0	0	0	0	0	0	0	0	0	0	0	l = Shipped - Damaged
														m = Shipping

2. How to calculate details by ourselves (From confusion matrix)

- **TP (True Positive)**
 - **Positive prediction & Positive sample → True**
 - Number within **Green Circle (4228)** as it located at cross of actual class (Horizontally) and classified class (Vertically) - arranged in diagonal line
- **FN (False Negative)**
 - **Negative prediction & Positive sample → False**
 - Sum of **Red Rectangle (27)** as it horizontally positive while vertically negative
 - → Orders was Cancelled, but being classified as shipped or pending (Any but other than Cancelled)
- **FP (False Positive)**
 - **Positive prediction & Negative sample → False**
 - Sum of **Orange Rectangle (25)** as it vertically positive while horizontally negative
 - → Being classified as Cancelled, but it actually shipped or pending (anything but other than Cancelled)
- **TN (True Negative)**
 - **Negative prediction & Negative sample → True**
 - Sum of the rest
(11370+754+4+2+9+52+58+1+31234+2+49+6+8+316+201+17+109=44192)
 - → Being classified as shipped or pending...(other than Cancelled), and it actually anything other than Cancelled
- **Accuracy**
 - Correctly classified instance among all
 - Sum of diagonal values divided by total sum

2. Summarized into data table - Take **Cancelled** as the positive value

Classifiers	Training Set size	TP Rate	FP Rate	Accuracy
J48 Tree	60%	99.40%	0.10%	97.9081%
J48 Tree	80%	99.20%	0%	97.9246%
Simple Cart (REP Tree)	60%	99.40%	99.40%	97.5017%
Simple Cart (REP Tree)	80%	99.00%	99.40%	97.5821%
Naive Bayes	60%	94.40%	96.10%	96.8435%
Naive Bayes	80%	94.00%	96.00%	96.8724%
K-Nearest Neighbors (IBK)	60%	47.30%	58.70%	89.5156%
K-Nearest Neighbors (IBK)	80%	49.10%	60.90%	90.3326%

Methodologies to most suitable classification

Classifications are evaluated by multiple scores which could be calculated or shown from confusion matrices in various classification results

Determining which score is better-representative to evaluate suitability **depends on nature of purpose**

There are basically three attributes to evaluate suitability of classification - **Recall, Precision and F-Measure**

Recall (TP Rate)

- How many Positive samples have been correctly predicted
- $R = \frac{TP}{TP+FN} = \frac{4228}{4228+27} = 0.9937$
- **Low Recall** (Large FN) means **Too Much negative predictions** (include lots of positive samples)
- **Suitable Case - Desire for High Recall**
 - As much as possible to figure Positive samples
 - **Overestimate** Positive samples
 - → **Medical diagnosis:** Severe consequence if Positive sample missed in diagnosis - This consequence negatively outweighs that healthy one mistakenly diagnosed into positive

Precision

- How many Positive predictions are actually Positive
- $P = \frac{TP}{TP+FP} = \frac{4228}{4228+25} = 0.9944$
- **Low Precision** (Large FP) means **Too Much positive predictions** (include lots of negative samples)
- **Suitable Case - Desire for High Precision**
 - Prevent Negative sample from being Positively predicted
 - **Underestimate** Positive samples
 - → **Spam mail detection**: To prevent Normal mails classified into Spam mail - This consequence negatively overweigh that spam mail failed to be detected

F-Measure

- Balanced score for Recall and Precision
- $F = 2 \cdot \frac{P \cdot R}{P+R}$
- **Suitable Case - Equally desired for High Precision and High Recall**

Case analysis

In our case, we want to predict the status of orders as to be cancelled or not in order to **minimize stock**, hence **save expenditure on storage**

Evaluate negative effect

1. **Non-Cancelled orders predicted as Cancelled**
 - Lead to **Lower Inventory** compared to demand
 - Possible **out-of-stock** or **Delaying**
2. **Cancelled orders predicted as Non-Cancelled**
 - Lead to **Overstock**
 - Possible **higher cost** in storage expenditure

Nature of our case - Selling clothes on Amazon

- Low Storage Cost per unit
- Availability to Clearance Promotion (Discount, Channel expansion)
- Low Inventory Loss Rate

- Existence of Popularity Cycle
- High user loyalty

Analyze acceptance of negative effects relying on the nature

1. Due to relatively **Low Storage Cost per unit, Low Inventory Loss Rate** and potential to commit **Clearance Promotion** to eliminate overstock, it is acceptable to have Lower estimation over cancelled ones
2. Importance to **build user loyalty**, therefore it necessary to **avoid delaying and stockout** circumstances
3. **Negatively comparison**
 - Overstock is less negative than out-of-stock and delaying
 - Overstock is more acceptable
 - Cancelled orders accept underestimate more then overestimate

Therefore, To underestimate Positive sample (Cancelled), we should avoid Negative samples being positively predicted, which means **Higher Precision** is more demanded

Conclusion

The dataset basically involves Amazon sales data of kinds of clothes in India, as we undergo case analysis that determine **Precision is better-representative score** to evaluate quality of classification.

Therefore, **Simple Cart (REP Tree)** with 80% as Training set is best model among all listed classifiers

Statistics visualized in Power BI

Classifier	Size	Recall	Precision	FP Rate	F-Measure	Accuracy
☐ Simple Cart (REP Tree)	80%	99.00%	99.50%	0.00%	99.40%	97.5821%
Simple Cart (REP Tree)	60%	99.40%	99.40%	0.10%	99.40%	97.5017%
☐ J48 Tree	60%	99.40%	99.40%	0.10%	99.40%	97.9081%
J48 Tree	80%	99.20%	99.20%	0.00%	99.40%	97.9246%
☐ Naive Bayes	80%	94.00%	98.00%	0.20%	96.00%	96.8724%
Naive Bayes	60%	94.40%	97.80%	0.20%	96.10%	96.8435%
☐ K-Nearest Neighbors (IBK)	80%	49.10%	80.20%	1.20%	60.90%	90.3326%
K-Nearest Neighbors (IBK)	60%	47.30%	77.30%	1.30%	58.70%	89.5156%

